

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 412 — Past Questions & Materials
by Renance

Course: PEE 412
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — PEE 412 PAST QUESTION 2018-2019

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF TECHNOLOGY

2018/2019 First Semester Examination

PNG 412: Basic Well Logging & Formation Evaluation

Answer ALL Questions. Time Allowed: 2 hrs. Date: 04/05/2019

QUESTIONS

- 1a) What are the three main uses of well logging?
- b) The best use of the micro-resistivity log is the detection of ?
- c) Mention two main, practical applications of Spontaneous Potential log.
- d) Determine hydrocarbon saturation in the sands, if any:

Sand R_{xo} R_{int} R_w R_e

A 80 15 0.1 10

B 120 18 1.2 8

e 160 25 11 9

Dd 180 30 0.6 5

- 2a) Explain the need for a log header and geothermal gradient.
- b) What is the implication of $MNOR < MINV$: High $MNOR$ and $MINV$?
- c) On a log heading, following are indicated at 70 °F:
 $R_m = 2.0 \text{ Ohm.m}$, $R_{me} = 1.3 \text{ Ohm.m}$ and $R_e = 3.5 \text{ Ohm.m}$
 Compute the corresponding values at 150 °F.
- 3a) Compare Well Logging, Well Testing and Core Analysis.
- b) Identify shale and productive zones on the given Fig. What are the tools?
- c) Mention the logs necessary for fresh water zones, assessing poor cementing job and identifying water entry into a gas well.
- c) The density log through an interesting interval of sandstone formation registered a bulk density of 2.32 gm/cc. Assuming a matrix density of 2.65 gm/cc, calculate the apparent porosity for $p_{mr} = 1.1 \text{ gnm/cc}$, $S_g = 40\%$ and $p_g = 0.2 \text{ gm/cc}$.
- d) What are the clues for false gas indication? How can fluid phases be identified using well logging?